Validity Is Not Difficulty: Supplementary Material

Anonymous CVPR submission

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A. SceneIR Schema and Coordinate Conventions

SceneIR records explicit $T_{\text{city}\leftarrow\text{ego}}$, $T_{\text{ego}\leftarrow\text{sensor}}$, and $T_{\text{city}\leftarrow\text{actor}}$ transforms. A return $x_{s,t}$ is fused in Actor coordinates as

$$x_i = T^{-1}_{\text{city}\leftarrow\text{actor},t} T_{\text{city}\leftarrow\text{ego},t} T_{\text{ego}\leftarrow\text{sensor},t} x_{s,t}. \quad (1)$$

The physical layer stores Actor-local surfels, known/unknown masks, normals, temporal and view support, matched rays, and source timestamps. Appearance primitives are referenced only through attachment provenance; Gaussian opacity is never read as collision occupancy. Actor identity, category, extent, trajectory, lifecycle, and analytic hazard fields are immutable across compiler actions.

Four actions. KEEP preserves supported query evidence. PROJECT moves an artifact only to its matched observed termination. COMPLETE adds a canonical surfel only when temporal and view support satisfy the frozen contract. UNKNOWN is the fallback when evidence is insufficient; it never becomes free.

B. Matched-Ray Physical Certificate

Automotive LiDAR is 2.5D ray evidence, not merely an unordered 3D point set [2]. For origin o, direction v, and observed termination d, the measured free interval is

$$\mathcal{F}(o, v, d) = \{o + sv \mid 0 \leq s < d - \epsilon\}. \quad (2)$$

A provenance-bearing PROJECT candidate is placed at the same ray's termination $o + dv$. It therefore cannot lie in $\mathcal{F}$. A candidate without an exact ray is marked UNKNOWN; nearest-surface projection is not substituted. This is the source of the paired zero-free-space result. It is a deterministic contract for matched rays, not a completeness proof for the whole Actor surface or a probabilistic sensor-noise guarantee.

The held-out targets are later frames excluded from both canonical fusion and query compilation. Depth, ray-termination, Chamfer, precision, and recall are computed against these target-only returns. This separation explains why zero paired free-space violation can coexist with the observed 17.19% per-Actor Chamfer tail.

Bidirectional visibility partition. For each held-out target ray, the nearest output within the frozen 0.20 m beam tube is classified as early, hit, or late; otherwise it is unmatched. In the reverse direction, an output point is supported at the first return, contradicted if it lies in the observed free interval, occluded if it is behind the return, and unknown when off ray. Occluded and unknown points are not penalized as free-space violations, preventing unseen Actor backs from being conflated with measurable contradictions.

The pooled result does not imply Actor-wise dominance: 64.04% add no early/visible contradiction and only 1.58% have absolute zero visible contradictions. All-Actor output attribution assigns 95.12% of new early rays to COMPLETE, but each costs against 14.96 new matched hits; target early returns fall by 9,439 overall. Moreover, only 13.49% of contradicted output points are completion points, so source-ray carving cannot selectively remove the observed tail. We retain completion and require the independent repair-or-abstain head for Actor-level authority.

The exact-once fresh cohort independently preserves this boundary: only 62.72% add no visible violation, 2.29% have exact zero visible contradiction, and 19.50% worsen in Chamfer. Among the latter, just 16.67% retain a non-inferior visibility F-score. Fresh confirmation therefore strengthens pooled three-dimensional evidence while continuing to reject a universal Actor certificate.

Frozen authority–certificate alignment. An exact identity join to the frozen P4 decisions shows why abstention must name its target. At 77.25% coverage, selected Chamfer-worsening falls from 19.50% to 14.60%, yet selected visible-failure risk is 36.39% versus 37.28% for always-repair. Its one-sided 95% Wilson upper bound is 40.40%, which remains above the baseline point risk; safe-visible AUROC is 0.533, and abstention captures 24.62% of visible failures. P4 authorizes Chamfer-nonworsening re-

Table 1. Visibility-conditioned target-only evidence on consumed and independently frozen fresh AV2 cohorts. Higher hit/precision/F-score is better; lower early termination is better. Each audit was frozen before its metrics were read.

Surface	Target-hit recall (%)	Early termination (%)	Visible precision (%)	Visibility F-score
Consumed clean query	49.69	3.395	99.61	0.663
Consumed corrupted baseline	55.61	3.589	98.38	0.711
Consumed HARP-3D	69.83	2.868	99.77	0.822
Fresh clean query	49.72	3.448	99.51	0.663
Fresh HARP-3D	68.99	2.931	99.68	0.815

pair but provides no finite-cohort confidence separation for bidirectional visibility.

Why a stricter provenance rule fails. If authority additionally requires zero COMPLETE outputs, every retained point has a KEEP or matched-hit PROJECT witness. Provenance-only coverage is 36.90%; after conjunction with P4 it is 23.71%. Dual visible failure falls to 20.16% (one-sided 95% upper 26.70%), but Chamfer worsening rises to 43.55% and hazardous-Actor coverage is only 3.52%. Completion-count unsafe-visible AUROC is 0.589. Thus observed provenance improves one physical axis while failing joint geometry and hazard authority; it cannot certify unseen views.

A targeted visibility head still fails joint authority. A frozen nuScenes-only head reaches safe-visible AUROC 0.753 on source test and 0.625 on AV2. Visibility-only AV2 selection has 11.63% risk at 8.22% coverage. Its P4 conjunction has 10.26% risk (95% upper 20.98%) but only 7.46% coverage, 35.90% Chamfer worsening, and 3.52% hazard coverage. The ranking transfers; the authority does not.

Defer-to-query changes the unit of risk. Every abstained Actor retains its query surface, so Actor and hazard semantics remain 100.0%. P4's population introduced-visible failure/composite Chamfer gain is 28.11%/ 0.083 m; P6-C is 30.78%/0.089 m; and P4-and-visibility is 0.76%/0.0012 m. The provenance policy is 4.78%/-0.0009 m and is strictly dominated by query-only. Thus low conditional risk can coexist with a worse fallback world.

Hazard-stratified accounting. For each frozen stratum, composite gain equals coverage times selected gain, while introduced-visible mass equals coverage times selected risk (maximum numerical residual 5.55×10^{-17}). Hazardous Actors are 27.15% of the cohort but account for 42.86% of P4 introduced failures and 56.42% of its Chamfer gain. P4 hazard risk is 47.37% versus 46.48% for always-repair; 45 of its 48 removed failures are instead clear. P6-C repairs 99.30% of hazardous Actors at 46.81% risk. Conversely, P4-and-visibility's zero observed hazardous failure covers only 3.52% of hazardous Actors. Thus hazard-state retention is not a hazard-stratum visibility certificate.

Fresh action provenance separates two failure channels. Across 1,435,391 target rays, the compiled surface adds 1.348% new early returns and yields 14.51 new hits per new early return. COMPLETE accounts for 94.70% of new early returns and 99.94% of new hits, whereas KEEP accounts for 95.68% of emitted-surface contradictions. Hazardous/clear new-early rates are 1.912%/0.980%; their COMPLETE hit-to-early ratios are 13.67/17.62. P4 and P6-C hazardous rates remain $1.006\times$/$1.001\times$ always-repair. Thus the selectors do not suppress the completion-driven hazardous-ray mechanism. PROJECT has zero emitted provenance because observed-hit projections collapse under KEEP-first voxel deduplication; this is not a causal zero-effect result.

C. Literal First-Return Correction and Monotone Boundary

The frozen AV2 attribution above selects the Euclidean-nearest output to each target point and then checks its projected depth. It is a target-aligned proximity diagnostic, not literal ray termination. Following raw-LiDAR occupancy evaluation [3], P20 instead takes the minimum positive projected depth among every surface point inside the same frozen beam tube. Query and compiled surfaces use the identical operator.

$$d_S(r) = \min\{s > 0 : o + sv \in S_\epsilon\}, \qquad \min \varnothing = \infty. \tag{3}$$

For any $S' \subseteq S$, Eq. 3 gives $d_{S'}(r) \geq d_S(r)$, so deletion cannot increase the early predicate or new-early relative to a fixed query. There is no corresponding guarantee for hits or symmetric Chamfer. Table 2 shows exactly this boundary on the consumed nuScenes source-development cohort: all three frozen deletion policies reduce hazardous early returns, yet all pay surface utility. P19 is most efficient, removing 131 hazardous events for 71 hits and 0.0919 mm mean Chamfer (1.845 events/hit; 1426.1 events/mm). These ratios are descriptive, not collision or road-safety guarantees.

D. Frozen Cohorts and Exact-Once Rules

After a quality read, scenes/logs cannot be replaced, failed cases remain in the denominator, and no threshold, radius, feature group, or checkpoint may change. Download/resource failures are reported separately from scientific failures. The fresh AV2 protocol is bound to the already frozen P6-C/P4 artifacts and is not changed by the rejected fresh nuScenes result.

The fresh AV2 read contains 523 Actors. P6-C/P4 repair AUROC is 67.62/65.48, coverage is 83.94/77.25%, and selective Chamfer is 0.178/0.184 m. This external reversal does not dominate P4: false repair is 12.43% versus 11.28%, and P6-C independently loses fresh-nuScenes AUROC and AURC. The complete evidence therefore retains P4 and rejects a domain-invariant sparsity-consistency claim.

E. Actor-Level Decision Intervals

For the frozen ReLU validity MLP, an input box $[\ell, u]$ is propagated through each affine layer using

$$\ell' = W^{+}\ell + W^{-}u + b, \quad u' = W^{+}u + W^{-}\ell + b, \tag{4}$$

followed by endpoint-wise ReLU. Here $W^{+} = \max(W, 0)$ and $W^{-} = \min(W, 0)$. The resulting logit interval is mapped through the sigmoid. An Actor is *robust-select* if its lower probability is above the frozen P4 threshold, *robust-abstain* if its upper probability is below threshold, and unresolved otherwise. This is deterministic interval-bound propagation over a specified box, related to robust-OOD certification [1]; it does not assert that the box describes an unseen sensor.

We fix radii $0.05/0.10/0.20$ in nuScenes-training standard deviations. The sensor-opportunity group contains point/surfel counts, temporal/view support, range, and frame count; the physical-surface group contains surface residuals, keep/unknown fractions, completion fraction, and compiled/query ratio. Their union is exactly the 13D validity input.

At radius 0.10, 33.33% of nominal nuScenes-test selections and 91.23% of consumed-AV2 selections are robust-select. Nevertheless, 10.76% of the latter are retained target false repairs. Query-point count is the largest one-feature interval source for 83.33%/95.11% of nuScenes/AV2 Actors. Thus a shifted model can be confidently and locally stable while wrong.

Table 2. Literal first-return source audit. Hazard early is the fraction of held-out hazardous-Actor rays newly terminated before the target. Chamfer and new hits use the same frozen surfaces; no policy is refit.

Surface policy	Completion coverage (%)	Hazard new-early (%)	Mean Chamfer (m)	New hits
Always-complete	100.00	8.742	0.194587	32,575
P17 ray-only	71.43	7.777	0.199411	29,191
P17R ray+Chamfer	88.96	8.230	0.195716	31,620
P19 one-slot veto	98.95	8.662	0.194679	32,504

Table 3. Cohort construction is metadata-only. No quality, model score, RGB appearance, or target metric is read before freezing.

Cohort	Size	Frozen rule
AV2 consumed external	30 logs	Sort all 150 Sensor-val UUIDs; take indices $0, 5, \dots, 145$. First 20 are quantitative and final 10 qualitative.
AV2 fresh exact-once	20 logs	Remove the consumed indices; from the 120-log complement take positions $0, 6, \dots, 114$.
nuScenes fresh exact-once	20 scenes	Exclude every P4 train/calibration/test scene; sort the remaining 106 available scenes by official index and take rounded equally spaced positions.

F. Failure Ledger and Claim Corrections

Table 4. Paper-facing failures are retained as boundaries rather than hidden by replacement experiments.

ID	Observation	Resulting boundary or correction
V7-F09	Aggregate geometry hides 17.19% Actor-level Chamfer worsening.	P3 is aggregate hard evidence; P4 adds repair-or-abstain and preserves all Actors.
V7-F11	AV2 validity attribution is dominated by sensor opportunity.	Zero hazard leakage does not imply domain-invariant validity.
V7-F13	Exact P4-train overlap is only two scenes/five Actors.	Reject joint fitting; audit the frozen physical/reliability interface only.
V7-F14	FP32 exceeds the analytic tolerance by 3.8×10^{-6}.	Recompute the same bound in FP64; do not loosen the tolerance or alter rows.
V7-F15	Fresh P6-C repair AUROC loses 3.50 points to P4.	Reject recovery despite higher operating-point coverage; P4 remains primary.
V7-F16	Freeze prose mislabeled the P346 held-out horizon.	Executable artifact is 2.5 s; correct documentation without rerunning.
V7-F17	47 AV2 false repairs remain robust-select at radius 0.10.	Network stability is not correctness, calibration, or a safety certificate.
V7-F18	Pooled ray physics improves, but consumed/fresh nonnew-visible rates are only 64.04/62.72%.	Retain aggregate C1 evidence and the completion gain–tail trade-off; reject a universal Actor certificate and do not tune the ray tolerance.
V7-F19	P6-C loses fresh-nuScenes ranking but improves fresh-AV2 AUROC while raising false repair.	Report the domain reversal; an external gate pass neither overrides the source rejection nor promotes P6-C over P4.
V7-F20	P4 visible risk is 36.39%, but its 40.40% upper bound exceeds the 37.28% baseline.	Repairability does not confidence-separate visibility risk.
V7-F21	Provenance lowers visible risk but raises Chamfer worsening to 43.55% and hazard-repair coverage falls to 3.52%.	Observed provenance is one-axis evidence, not joint authority.
V7-F22	The source visibility head transfers ranking, but dual coverage/gain are negligible and hazard-repair coverage is 3.52%.	Reject the head family rather than sweep target thresholds.
V7-F23	Provenance defer has -0.0009 m composite gain with nonzero introduced failure.	Query-only strictly dominates it; conditional risk is not fallback-system utility.
V7-F29	Literal first-return raises baseline hazard exposure to 8.742%; P17/P17R/P19 all lower it but worsen Chamfer.	Point deletion has a certified early-return direction, not hit, Chamfer, collision, or policy safety.

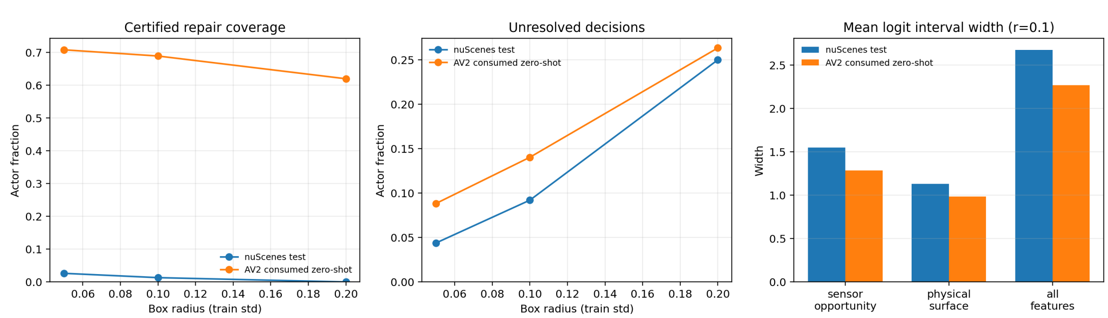

Figure 1. Frozen P7-C interval audit. Left/center: robust-select coverage and unresolved fraction across pre-registered radii. Right: sensor-opportunity boxes induce wider logit intervals than physical-surface boxes. The high AV2 certified coverage does not erase the 47 stable false repairs.

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G. Frozen AV2 Camera-Evidence Gallery

The following ten cases are the actor-rank-0 case from each qualitative log in the pre-registered order, not a metric or appearance ranking. Every panel shows synchronized RGB, the paired synthetic artifact probe, the four-action physical output, and sparse camera depth. The full package contains 30 panels and 30 before/after videos (three lexically first eligible Actors per log). Videos diagnose temporal behavior but are not photorealistic renders.

Figure 2. Frozen case q00, hazardous Actor 1f197615, ring-front-center; 251 visible query returns and 2,287 sparse-depth points in crop.

Figure 3. Frozen case q01, non-hazardous Actor 0e0562b6, ring-rear-left; 137 visible query returns and 1,770 sparse-depth points.

Figure 4. Frozen case q02, non-hazardous Actor 08fc423e, ring-side-right; 85 visible query returns and 2,476 sparse-depth points.

Figure 5. Frozen case q03, hazardous Actor 05cccb41, ring-side-left; 2,456 visible query returns and 17,286 sparse-depth points.

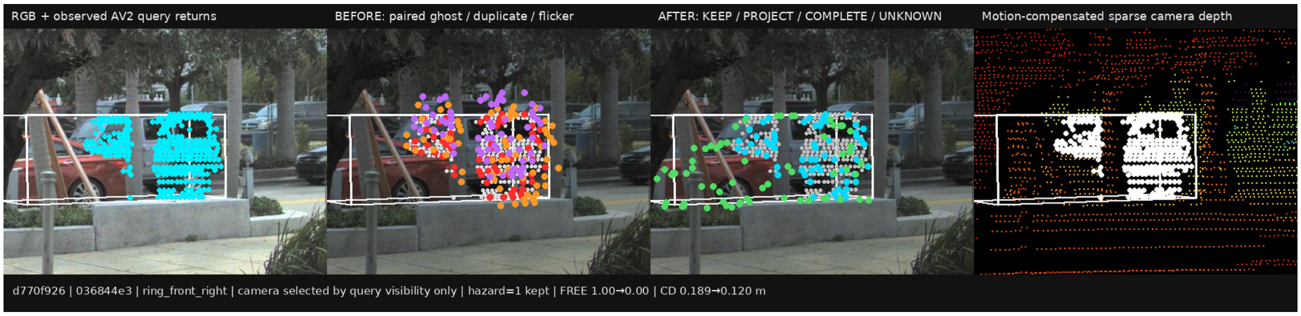

Figure 6. Frozen case q04, hazardous Actor 036844e3, ring-front-right; 292 visible query returns and 2,358 sparse-depth points.

Figure 7. Frozen case q05, hazardous far/sparse Actor 07695d24, ring-front-center; all 14 query returns remain visible.

Figure 8. Frozen case q06, non-hazardous Actor 08515e0d, ring-front-right; 79 visible query returns and 2,757 sparse-depth points.

Figure 9. Frozen case q07, non-hazardous Actor 076c9103, ring-front-center. This unfiltered case exposes Chamfer worsening and is retained.

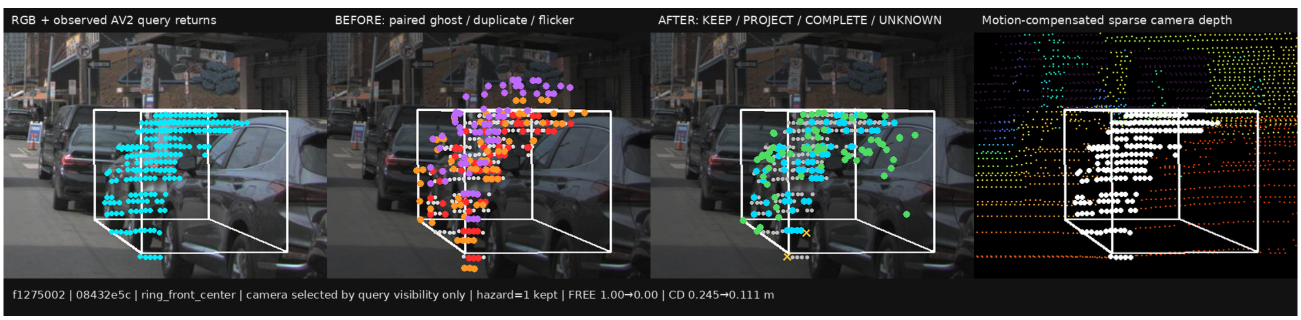

Figure 10. Frozen case q08, hazardous Actor 08432e5c, ring-front-center; 231 visible query returns and 2,101 sparse-depth points.

Figure 11. Frozen case q09, non-hazardous Actor 6d55ff95, ring-front-center; 38 visible query returns and 1,630 sparse-depth points.

H. Resource and Reproducibility Boundary

All V7 scientific runs use one RTX 3090. The largest paper-facing GPU allocation is below 1.5 GiB, while P7-C uses 9.2 MiB and 1.37 s. Download and preprocessing are sequential on the low-memory host. Artifacts are identified by task/run paths and immutable cohort/config files; the protocol intentionally does not add hashes, checksums, fingerprints, repeated smoke matrices, or post-hoc quality gates.

References

[1] Julian Bitterwolf, Alexander Meinke, and Matthias Hein. Certifiably adversarially robust detection of out-of-distribution data. In *Advances in Neural Information Processing Systems*, 2020. 3

[2] Peiyun Hu, Jason Ziglar, David Held, and Deva Ramanan. What you see is what you get: Exploiting visibility for 3d object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11001–11009, 2020. 1

[3] Jonas K”alble, Sascha Wirges, Maxim Tatarchenko, and Eddy Ilg. Accurate training data for occupancy map prediction in automated driving using evidence theory. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5281–5290, 2024. 2